Jump Modeling with Dirichlet Process Mixture

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Highlights of Contributions

- Flexible modeling of price jump size distribution through infinite mixtures - **Dirichlet process mixture** (DPM);
- Recover the **empirical distribution** of jump size;
- Provide **better predictive density** of return distributions;
- Provide a model generated implied volatility (IV) smile that fits better to the empirical **IV smile**.

Motivation

1. Jumps (extreme and sudden changes) matter in terms of modeling stylized facts of **return** distribution, pricing **options** and fitting to **implied volatility** smile.
2. Conventional parametric model often assumes jump size **follows one specific distribution**, which is not flexible enough for jump modeling in real life.
3. With observing returns only, it is difficult to distinguish latent volatility and price jumps. Possible solutions: **high frequency volatility measure** to infer the movements of volatility.

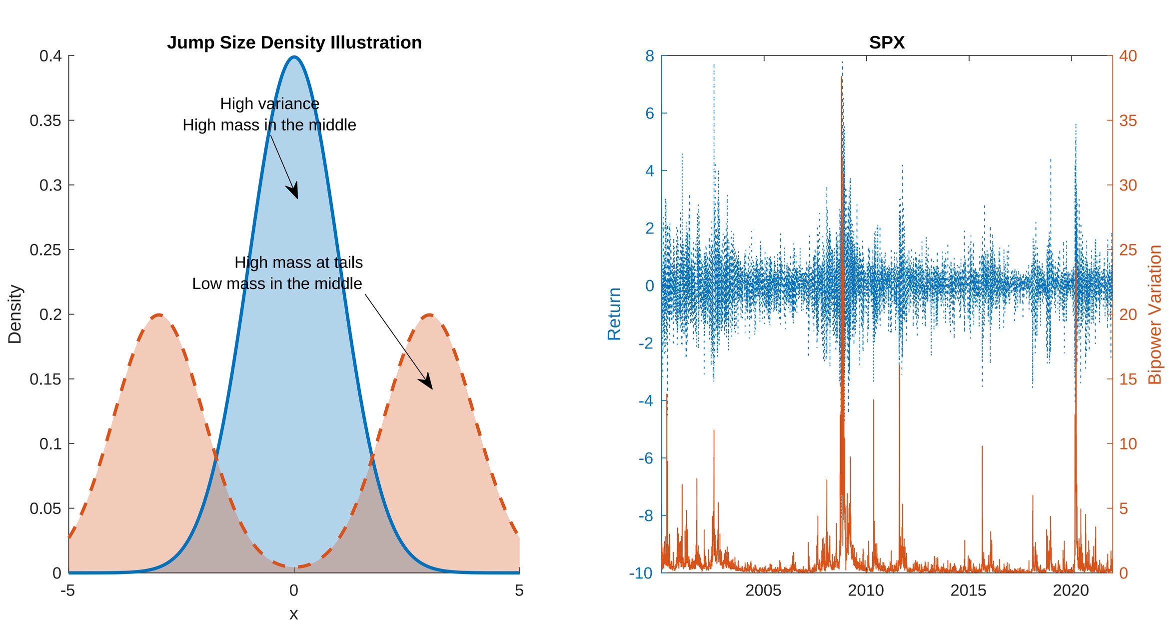

Figure 1. Left panel is an illustration of using unimodal and bimodal normal distribution for jump size modeling, which is generated randomly. Right panel is the empirical daily return and the bipower variation of SPX where bipower variation indicates the movement of latent volatility.

Jump Modelling with DPM

Let r_t be the daily log return. Let h_t be the latent log variance, ΔN_t be the jump occurrence and Z_t be the jump size. Our proposed model follows:

$$\begin{aligned} r_t &= \exp(h_t/2)\epsilon_t^r + \Delta N_t Z_t, \\ h_{t+1} &= \phi_0 + \phi_1 h_t + \sigma_h \epsilon_t^h, \\ (\epsilon_t^r, \epsilon_t^h)^\top &\sim N\left(0, \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}\right) \\ \Delta N_t &\sim \text{Bernoulli}(\delta) \end{aligned} \tag{1}$$

$\mathcal{M}_1$: Static DPM jump model

$$Z_t \sim \sum_{k=1}^{\infty} w_k N(\mu_k, \sigma_k^2), \tag{2}$$

$\mathcal{M}_2$: Dynamic DPM jump model

$$Z_t \sim \sum_{k=1}^{\infty} w_k N(\mu_k (\bar{BV}_t)^{1/2}, \sigma_k^2), \tag{3}$$

where

$$\begin{aligned} w_k &= \nu_k \prod_{l=1}^{k-1}(1-\nu_l),\ \nu_k \sim \text{Beta}(1,\alpha), \\ BV_t &= \frac{\pi}{2}\left(\frac{N}{N-1}\right)\sum_{i=2}^{N}|r_{t_i}||r_{t_{i+1}}|,\ \bar{BV}_t = \frac{1}{n}\sum_{m=1}^{n} BV_{t-m} \end{aligned} \tag{4}$$

for which r_{t_i} is the ith of total N observed returns between trading day t and $t+1$.

$\mathcal{M}_3$: Benchmark parametric model

$$Z_t \sim N(\mu_z, \sigma_z^2), \tag{5}$$

Results Discussion

To illustrate the usage of our proposed model, we focus on the following aspects of the interested problems and demonstrate the performance of our model through simulation, empirical analysis and applications in IV smile.

What we are interested in:

- How DPM jump models perform under different data generating process (DGP), i.e. low (high) volatile time periods ⇒ "**Simulation**":
- Can DPM jump models provide better predictive density of returns ⇒ "**Empirical Analysis**"
- Can DPM jump model generated IV smile fits to the empirical one better? ⇒ "**Empirical Application**"

Sampling Mechanics

A. How to deal with the infinite sum – Slice Sampler:

Introducing an auxiliary variable u_t who is uniformly distributed conditional on $\boldsymbol{w}$, then we have

$$\begin{aligned} p(Z_t, u_t|\cdot) &\propto \sum_{k=1}^{\infty}\frac{1}{w_k} I(u_t < w_k) w_k N(Z_t|\mu_k, \sigma_k^2), \\ &\propto \sum_{k\in A_t} N(Z_t|\mu_k, \sigma_k^2), \end{aligned} \tag{6}$$

where $A_t = \{k : u_t < w_k\}$. Note here A_t is a **finite** set where the condition is satisfied.

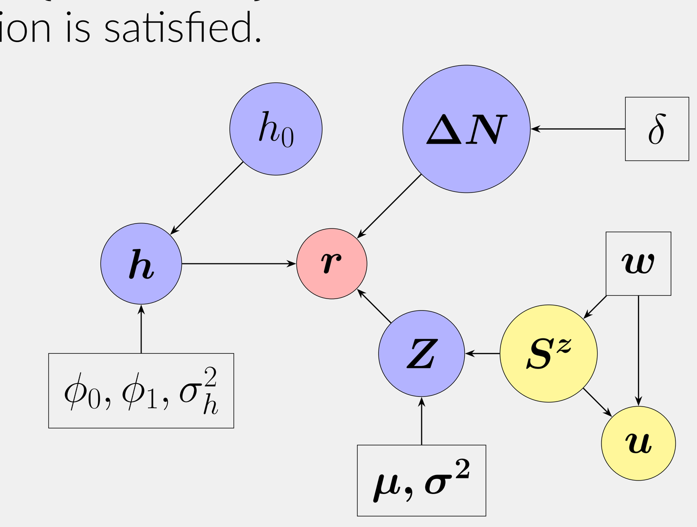

Figure 2. DAGs of the DPM jump model structure. Red circles represents observed data, purple circles are latent variables. Parameters are represented as rectangular. Yellow circles represents the auxiliary variables where $\boldsymbol{u}$ is introduced for slice sampler and $\boldsymbol{S^z}$ is an indicator functions for components.

B. How we deal with the non-linear state-space model – "7 component mixture":

$$\begin{aligned} r_t^* &= r_t - \Delta N_t Z_t, \\ \log(r_t^{*2} + 0.001) &= h_t + \sum_{i=1}^{7} q_i N(m_i - 1.2704, \nu_i^2), \end{aligned} \tag{7}$$

We approximate the distribution of $\log(\epsilon_t^{r2})$ as a mixture of seven component with pre-selected mean m_i and variance ν_i^2 in the literature.

Simulation Analysis

For simulation, we set $\rho = 0$ and define the true DGP as following and generate observations with different δ and μ_z from:

$$\begin{aligned} r_t &= \exp(h_t/2)\epsilon_t^r + \Delta N_t Z_t, \\ h_t &= -0.06 + 0.94 h_{t-1} + 0.38\epsilon_t^h, \\ \log(BV_t) &= h_t + 0.4\epsilon_t^{BV} \\ \bar{BV}_t &= \frac{1}{5}\sum_{m=1}^{5} BV_{t-m} \\ \Delta N_t &\sim \text{Bernoulli}(\delta), \\ Z_t &\sim 0.6 \times N(-\mu_z e^{h_t/2}, 0.8) + 0.4 \times N(\mu_z e^{h_t/2}, 0.8). \end{aligned} \tag{8}$$

Robustness Check of DPM Model Performance:

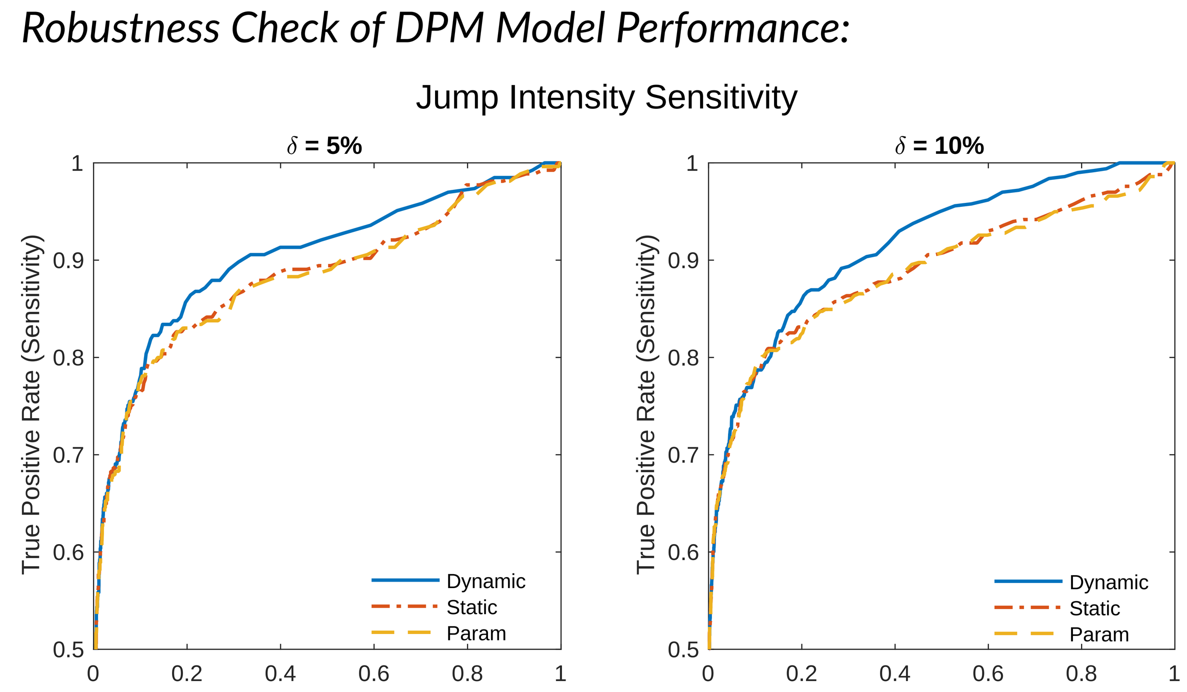

Figure 3. Receiver Operating Characteristics (ROC) curves for **jump intensity analysis**. The true jump size mean is fixed as $3e^{h_t/2}$ but the jump intensity is at 5% and 10%. The blue solid line represents the ROC curve of dynamic SV-DPM jump model ($\mathcal{M}_1$) the red dot-dashed line represents the ROC curve of static SV-DPM jump model ($\mathcal{M}_2$) and the yellow dashed line is the ROC curve of parametric model ($\mathcal{M}_3$). The curve is closer to the top left corner, the bigger the power.

Area under the ROC curve (AUC)			
	Dynamic DPM	Static DPM	Parametric SVJ
A: Jump Intensity Analysis ($\mu_z = 2$)			
$\delta = 1\%$	0.8960	0.8791	**0.9043**
$\delta = 5\%$	**0.9054**	0.8883	0.8819
$\delta = 10\%$	**0.9004**	0.8816	0.8788
$\delta = 15\%$	**0.9208**	0.9018	0.9018
B: Jump Size Analysis ($\delta = 10\%$)			
$\mu_z = 2$	0.7966	0.7932	**0.7974**
$\mu_z = 2.5$	**0.8491**	0.8270	0.8287
$\mu_z = 3$	**0.9004**	0.8816	0.8788
$\mu_z = 3.5$	**0.9363**	0.9160	0.9171

Table 1. **Area under the ROC curve (AUC)** for both jump intensity analysis and jump size analysis under simulation. The larger AUC, the more accurate the model is in terms of detecting true jumps. The bold numbers are the maximum of each cases.

Empirical Study Key Findings

- Dynamic DPM model provide **more separated jump size density** when the market is volatile.
- Dynamic DPM model can provide **better predictive density** of returns than parametric SVJ model. We observe gains mostly on the prediction of 10% and 90% quantiles and of the whole distribution. Significant gains are observed especially during very volatile time periods.
- Generate IV smile that **fits to the empirical IV smile better** for in-the-money options.

Empirical Analysis

The model implied jump size density over time:

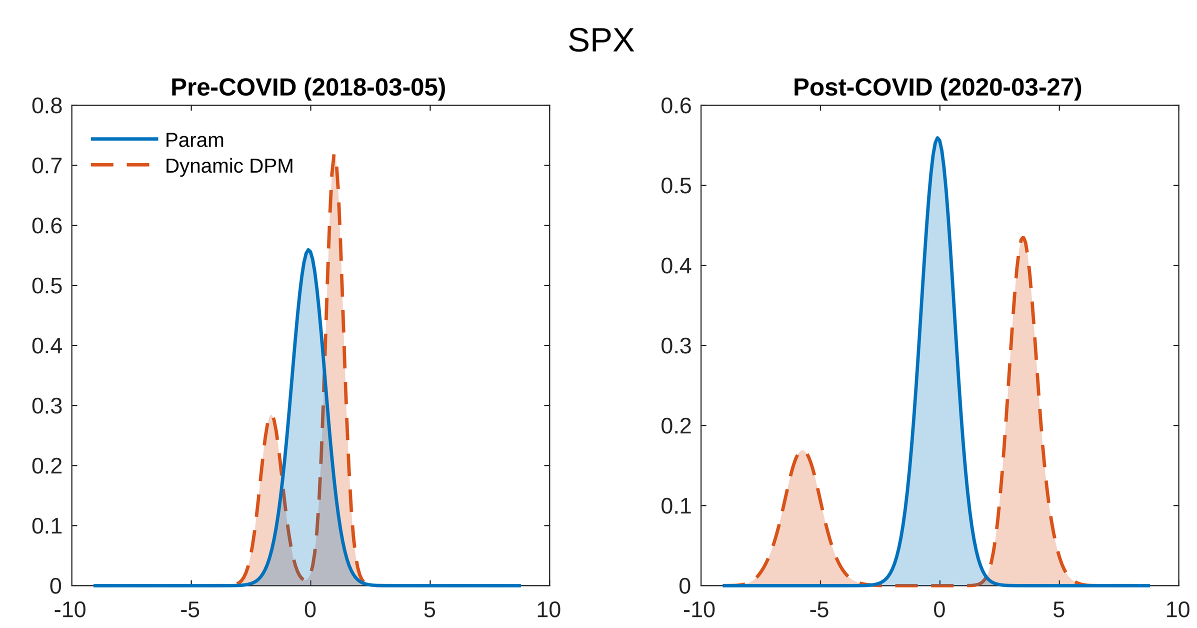

Figure 4. The model implied jump size density pre-COVID and post-COVID for SPX. The blue solid line represents the model implied jump size density for parametric model. The red dashed line represents the model implied jump size density for dynamic DPM model.

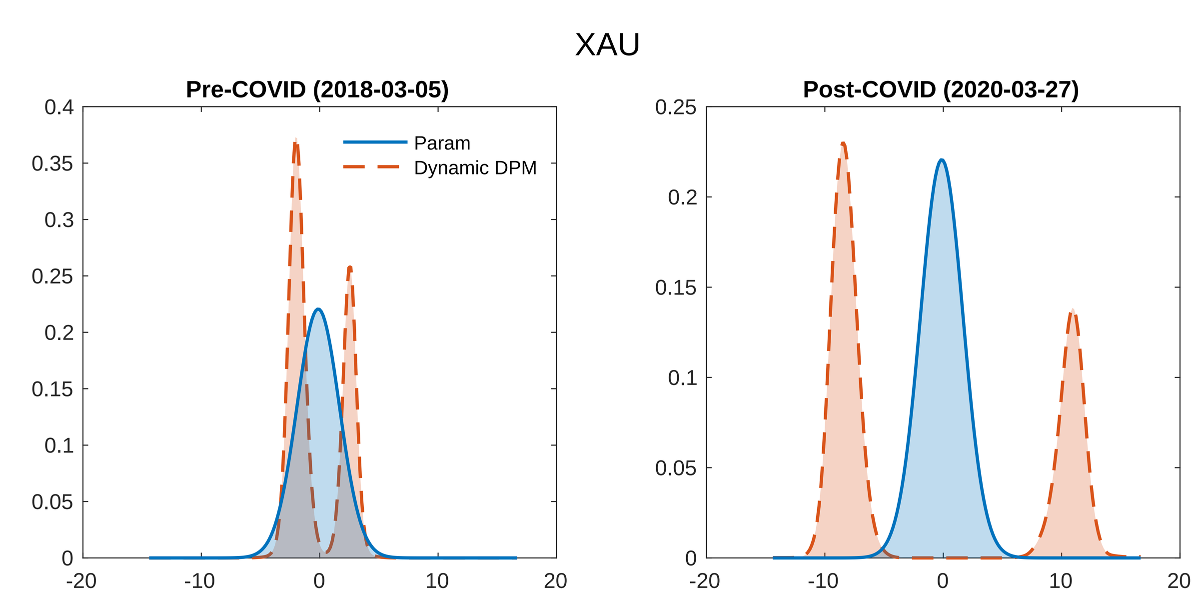

Figure 5. The model implied jump size density pre-COVID and post-COVID for XAU. The blue solid line represents the model implied jump size density for parametric model. The red dashed line represents the model implied jump size density for dynamic DPM model.

Evaluation of predictive density by cumulative score difference:

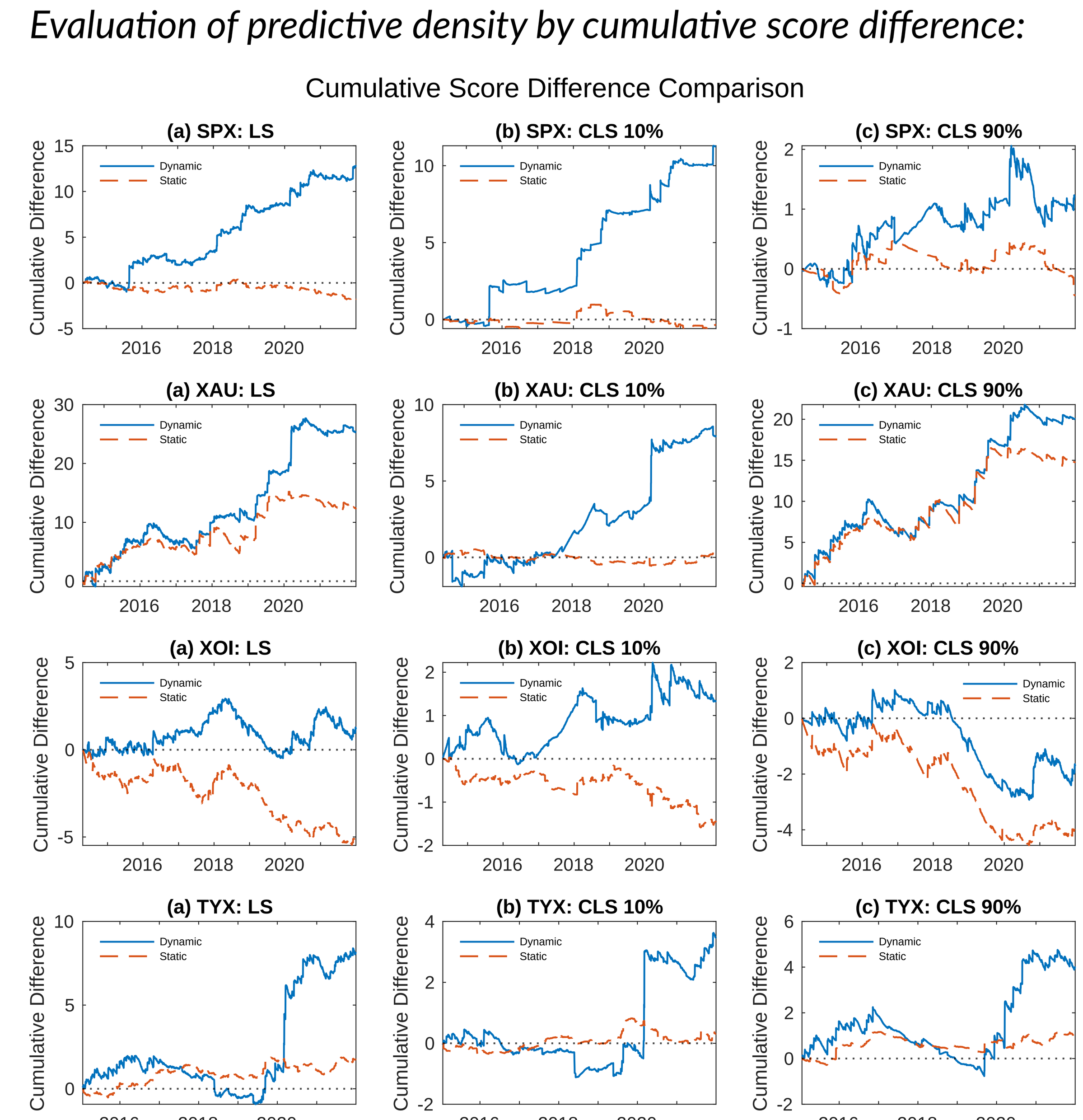

Figure 6. Cumulative score difference between DPM models and parametric SVJ model when the leverage effect is not considered. The blue solid line is the cumulative score difference between dynamic DPM and parametric SVJ model and the red dashed line shows the cumulative score difference between static DPM and parametric SVJ model. LS, CLS of 10% and 90% quantile are documented. If the cumulative score difference is above zero, then it shows that the DPM model outperforms the parametric model.

Verification of DPM jump models via IV smile: SPX normalized IV

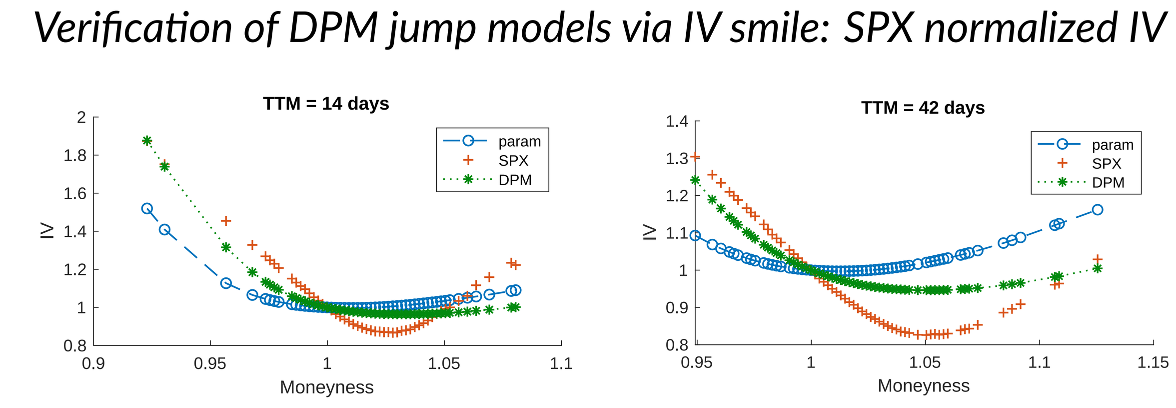

Figure 7. Normalized empirical IV smile with the time-to-maturity being 14 and 42 days, plotted together with the model generated IV from dynamic DPM model and parametric SVJ model. Leverage is not considered here yet.

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